



Chapter 3

Introduction to HTML5:

Part 2

Internet & World Wide Web
How to Program, 5/e

<ftp://111webdesign:webdesign@120.114.190.22:4000>

(課程範例檔案)





OBJECTIVES

In this chapter you'll:

- Build a form using the new HTML5 **input** types.
- Specify an **input** element in a form as the one that should receive the focus by default.
- Use self-validating **input** elements.
- Specify temporary **placeholder** text in various **input** elements
- Use **autocomplete input** elements that help users re-enter text that they've previously entered in a form.
- Use a **datalist** to specify a list of values that can be entered in an **input** element and to autocomplete entries as the user types.
- Use HTML5's new page-structure elements to delineate parts of a page, including headers, sections, figures, articles, footers and more.

3.1 New HTML5 Form **input** Types

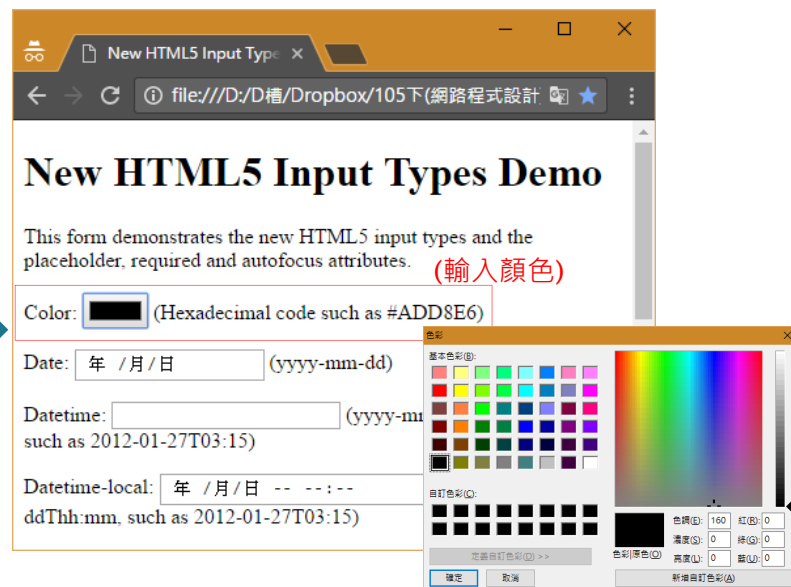
- ▶ HTML5's **new form input types**. (支援新輸入型態)
 - **not** universally supported by all browsers.

<input type = "color" ... /> 輸入顏色

- ▶ enables the user to enter a **color**.
 - 註：供使用者輸入顏色，ex: 詢問喜好顏色, 幸運色,...
- ▶ Most browsers display the **color input type** as a **text field** in which the user can enter a **hexadecamal code** (16進位碼) or a **color name** (ex: **red**, **green**, **blue**).
 - Some browser (e.g., **chrome**) can display a **color picker** (著色盤).



```
16 <form method = "post" action = "http://www.deitel.com">
17 <p>
18 <label>Color:
19 <input type = "color" autofocus />
20 (Hexadecimal code such as #ADD8E6)
21 </label>
22 </p>
23 <p>
24 <label>Date:
25 <input type = "date" />
26 (yyyy-mm-dd)
27 </label>
28 </p>
29 <p>
30 <label>Datetime:
31 <input type = "datetime" />
32 (yyyy-mm-ddTThh:mm+ff:gg, such as 2012-01-27T03:15)
33 </label>
34 </p>
35 </form>
36
```



(chrome會跳出調色盤)

3.1.1 input Type color

autofocus (屬性：自動游標定位點)

- ▶ an optional (選擇性/非必要) attribute that can be used in only one input element on a form
 - 註：游標落在指定輸入位置，方便使用者輸入

validation 相關驗證屬性

- ▶ New HTML 5's input types are **self validating** (供自行驗證) on the client side
 - 1) 免去使用複雜的JavaScript碼 (過去都是靠javascript來作驗證)
 - 2) 避免送出不合法的資料 (免去server不必要的計算)
 - 3) 有效減少往返Internet流量
- ▶ The **server** should still **validate** all user input.
 - 註：Client端僅驗證資料格式 (ex: 是否符合大小寫、數字)，而Server端才是驗證資料內容 (ex: 帳號、密碼是否正確)

3.1.1 input Type color (cont.)

- ▶ The following table shows the new HTML5 input types and provides examples of the proper formats for each type of data to be valid (合法格式).

- 註：常用輸入型別

input type	Format
color	Hexadecimal code (#16進位碼)
date	yyyy-mm-dd (年-月-日)
datetime	yyyy-mm-dd (年-月-日 時:分:秒 時區)
datetime-local	yyyy-mm-ddThh:mm (年-月-日 時:分)
month	yyyy-mm (年-月)
number	Any numerical value (數字)
email	name@domain.com (email)
url	http://www.domain-name.com (網址)
time	hh:mm (時:分)
week	yyyy-Wnn (年-週, ex: 2025-W01)

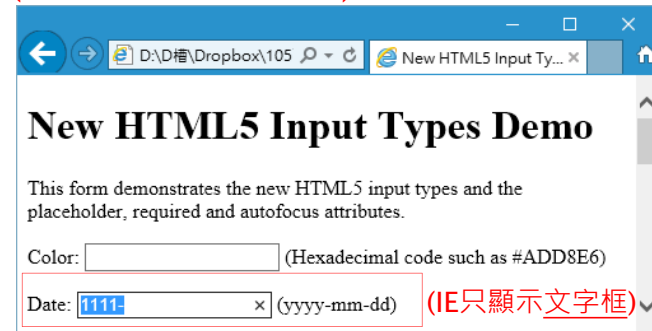
3.1.1 input Type color (cont.)

formnovalidate ... 跳過不檢查

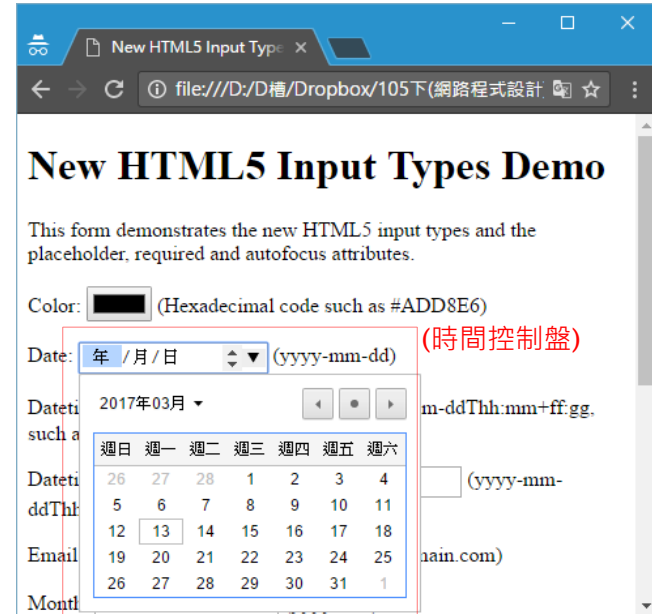
- ▶ To **bypass** (繞過) validation, we can use the **formnovalidate** attribute to input type submit:

```
<input type = "submit" value = "Submit"
formnovalidate />
```

IE (其他瀏覽器，若不支援):



Chrome瀏覽器:



<input type="date" ... /> 輸入日期 (01~12) (01~31)

- ▶ To enter a **date** in the form **yyyy-mm-dd**
 - 註：確保輸入格式一致，ex：用於統計或圖表.
- ▶ Note:
 - **Firefox** and **Internet Explorer** display a text field in which a user can enter a date such as **2025-01-27**.
 - 註：**Firefox**和**IE**不支援，只顯示文字框需自行輸入
 - **Chrome** and **Safari** display a **spinner control** (控制盤)
 - a **text field** with an **up-down** arrow (可上下選擇) on the right side
 - allowing the user to **select a date** by clicking the up or down arrow.

3.1.3 input Type datetime

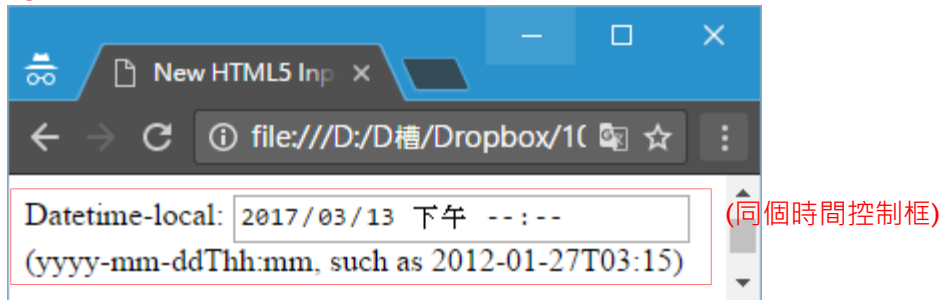
`<input type = "datetime" />` 輸入日期 (包含: 時/分/秒/時區)

- ▶ To enables the user to enter a **date** (**year, month, day**), **time** (**hour, minute, second, fraction of a second**) and the **time zone** set to **UTC** (Coordinated Universal Time, 法語: Temps Universel Coordonné, 簡稱: 世界時間標準).
 - Most of the browsers render datetime as a text field; 註: 若瀏覽器都不支援, 只顯示"文字框"

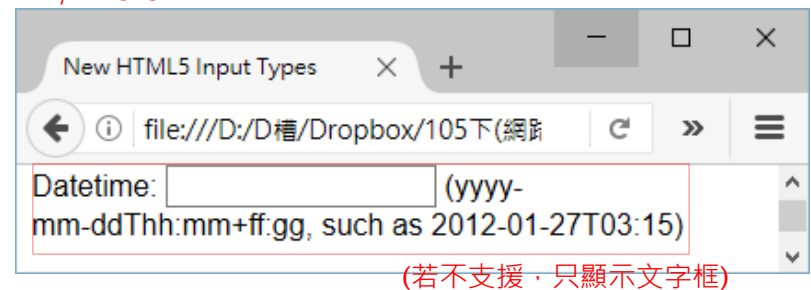
`<input type = "datetime-local" />` 輸入日期(包含: 時/分/秒)

- 差異: 無時區

Chrome:



IE/Firefox:



- ▶ 註: 其住瀏覽器, 像: Internet Explorer, Firefox and Safari 只顯示文字框.

3.1.5 input Type email

`<input type = “email” ... />` 輸入信箱 (Ex: 帳號@domain)

- ▶ To enter an e-mail address or a list of e-mail addresses separated by commas (分號“;”).
 - 註1): 可輸入一個或多個信箱(用分號;隔開)
 - 註2): 所有瀏覽器都以文字框作輸入
- ▶ If the user enters an *invalid* e-mail address (ex: 不含@等)
 - After clicking the Submit button, a callout (對話框) asking the user to enter a valid e-mail.
 - 註: 若輸入不完整(或不合法), 會跳出對話框要求更正。

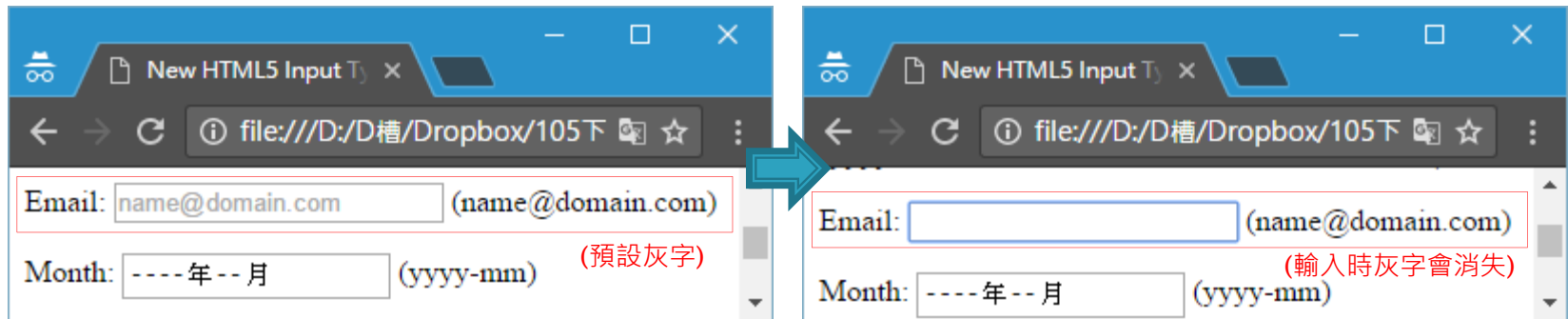
The screenshot shows a web browser window titled 'New HTML5 Inp'. The address bar shows 'file:///D:/D槽/Dropbox/1C'. The form has three input fields: 'Email: 11' (with a tooltip warning about missing '@'), 'Month: ----年--' (with a dropdown arrow), and 'Number: 4' (with a note '(Enter a number between 0 and 7)').

- *HTML5 **does not** check whether an e-mail address actually **exists** (是否為有效信箱)
 - it just **validates** that the e-mail address is in the *proper format*.
 - 註: HTML網頁不會檢查email是否能寄達信箱, 只驗證格式

3.1.5 input Type email (cont.)

placeholder="..." (屬性：預設文字)

- ▶ To place **temporary text** (暫存文字) in a text field.
- ▶ Generally, **placeholder** text is *light gray* (淺灰色) and provides **an example** of the text and/or text format the user should enter (Fig. 3.8).



- ▶ It's not "**submitted**" when the user clicks the Submit button.

註：灰色字不會送出

3.1.5 input Type email (cont.)

- ▶ HTML5 supports **placeholder** text for only six input types—**text**, **search**, **url**, **tel**, **email** and **password**.
 - 註：只有6種input types可以使用**placeholder**屬性(預設文字)
- ▶ **<... required ...>** (必填屬性)
 - ▶ **Must enter a value** before submitting the form.
 - ▶ **All of the input types** are **supported**
 - 註：所有input type都支援.
 - Example: the user **must** enter an e-mail address.

Browser window: New HTML5 Input Ty x

Address bar: file:///D:/D槽/Dropbox/105下

Form fields:

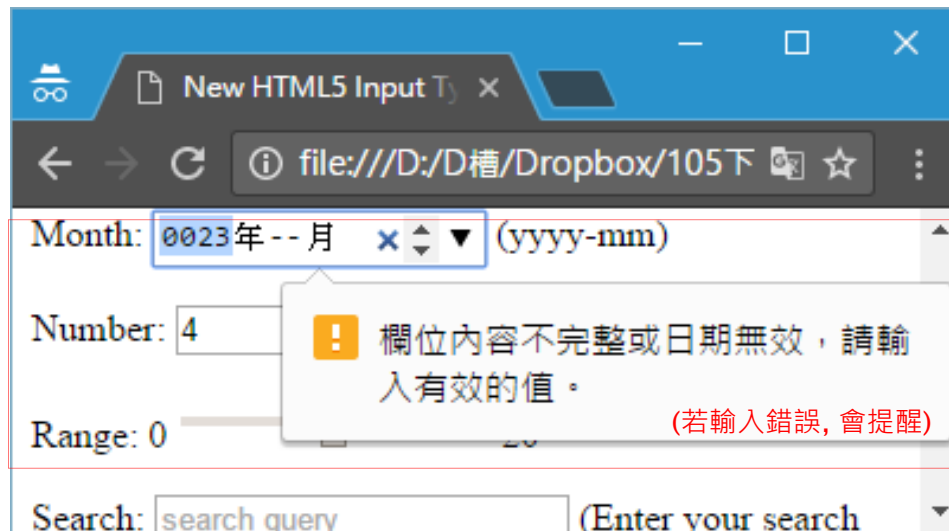
- Email: (name@domain.com)
- Month: ! 請填寫這個欄位。(若未填,則會提醒)
- Number: (Enter a number between 0 and 7)
- Range: 0 20
- Search: (Enter your search query here.)
- Tel: (###) ###-####

3.1.6 `input` Type `month`

`<input type = "month" />` 輸入年月(yyyy-mm)

- ▶ To enter a `year` and `month` in the format `yyyy-mm`, such as `2025-01`.

- 註：若輸入 (e.g., January 2025) 會跳出提醒框。

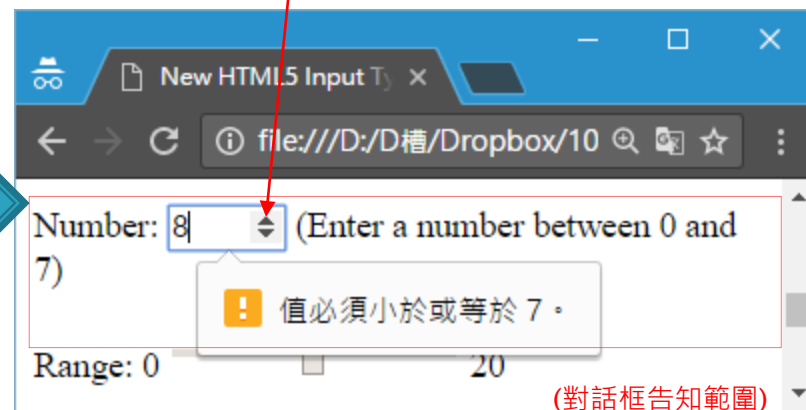
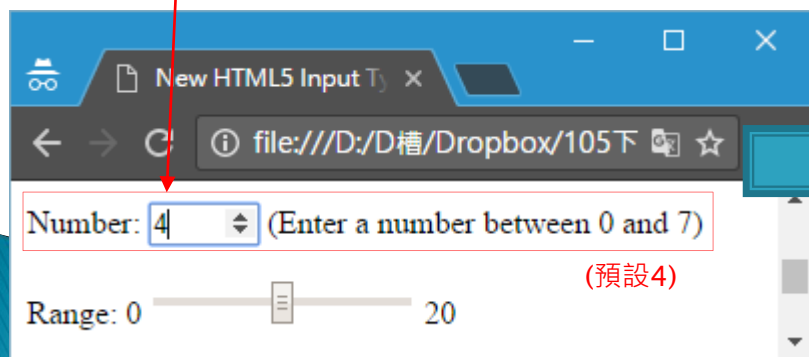


3.1.7 input Type number

`<input type = “number” ... />` 輸入數字 (如：年齡、幸運數字)

► To enter a **numerical value**

- Mobile (行動裝置) browsers: 顯示 **numeric keypad** (數字鍵盤).
- Internet Explorer, Firefox and Safari: 只顯示 “文字框”.
- Chrome, Opera : 顯示 **spinner control** (上下鍵控制盤).
- **min=“...”** : sets the **minimum valid number** (限定最小值, ex:年齡).
- **max=“...”** : sets the **maximum valid number** (限定最大值).
- **step=“...”** : the **increment** of the numbers (每次加/減數值).
- **value=“...”** : the **initial value** (預設值).
- If entering an **invalid value**, a **callout** pointing to the number input element will instruct the user to enter a valid value.
 - 註：若輸入錯誤，對話框會提醒範圍



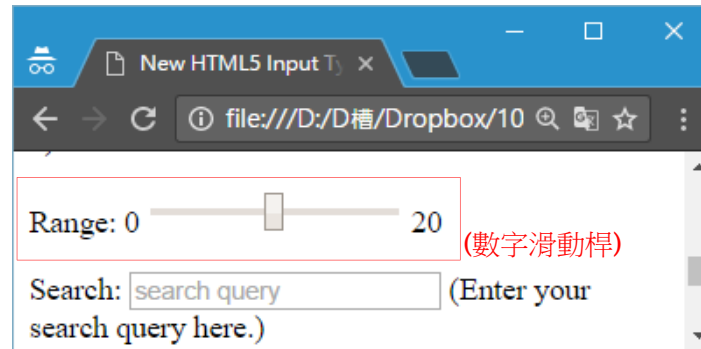
3.1.8 input Type range

(避免使用者輸入不合理數字)

`<input type = "range" ... />` 數字滑動桿(輸入數字)

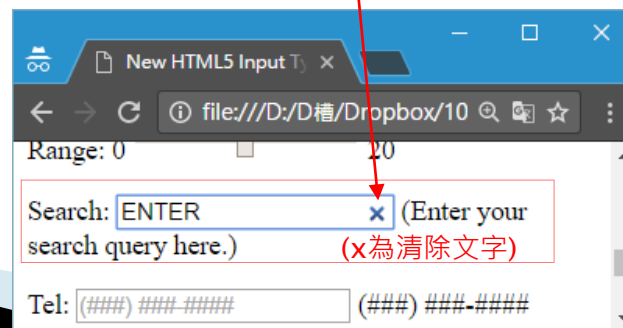
- Appears as a *slider control* (滑動控制) in Chrome, Safari and Opera.

- `min="..."` : 限定最小值.
- `max="..."` : 限定最大值.
- `value="..."` : 預設值.
- 註：避免發生輸入錯誤的情況 (無法拖出桿子外)



`<input type = "search" ... />` 提供搜尋

- Provides a *search field* for entering a **query**.
 - This input element is *functionally equivalent* (等同於) to an input of type **text** (文字輸入).
 - Chrome and Safari display an "x" that can be clicked to **clear** the field.
 - 註：Chrome、Safari瀏覽器的文字框有一個"x"，功用為清除文字。



3.1.10 input Type tel

(ex: 國際碼、區碼...)

`<input type = "tel" ... />` 輸入電話號碼

- ▶ To enter a **telephone number**
 - mobile browsers: 顯示 **keypad**(數字鍵盤).
 - all of the browsers : 顯示 **text field** (文字框).
 - 註: HTML5 **不會驗證**電話是否會通

`pattern="..."` 電話號碼規則 (ex: (06)2606-123)

- ▶ To uses a **regular expression** (正規表達式) to **limit** the number is in the right format:
 - Ex: (06)2606-123 → `\(\d{2}\)\d{4}-\d{3}`
- ▶ When the number is in the **wrong format**, a **callout** requests the proper format.

Tel: (###) ###-#### (###) ###-####
(文字框)

Time: -- --:-- (hh:mm)

URL: http://www.domainname.cc
(http://www.domainname.com)



Tel: (03) 571-2121 (###) ###-####

Time: -- --:--

URL: http://www.domainname.cc
(http://www.domainname.com)

(錯誤提醒: 區域碼應為3碼)

3.1.11 input Type time

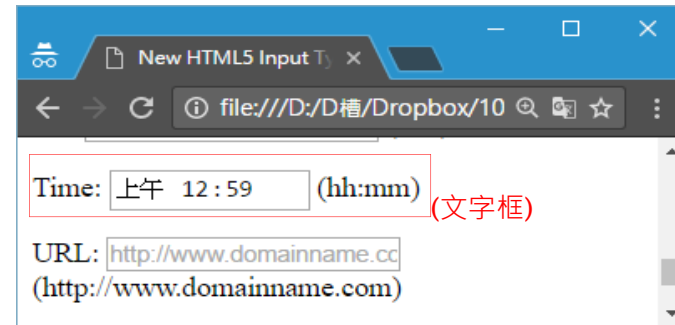
(只想調查時間，ex: 晚餐時間)

(ps. 先前datetime: 日期+時間)

`<input type = "time" />` 輸入時間 (hh:mm)

► To enter an **hour**, and **minute**.

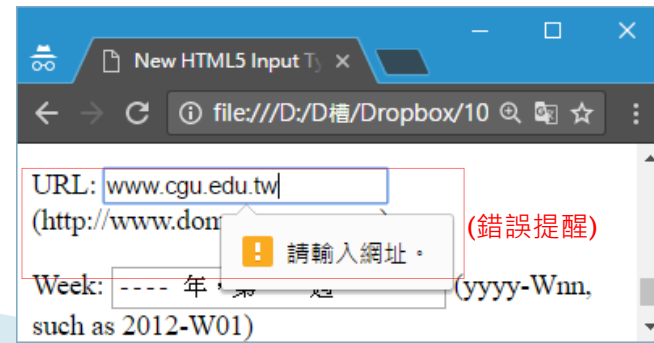
- A time must have **two digits** (2位數) representing the **hour** (01~12), followed by a **colon** (冒號分隔小時和分鐘":") and **two digits** representing the **minute** (00~59).
 - Ex: (hh:mm) 小時、分鐘都是2位數



`<input type = "url" ... />` 輸入網址

- Displayed as a **text field** (文字框), and the proper format is **http://www.deitel.com** (須包含: **http**協定和完整URL).
- **improperly formatted** URL (ex: **www.deitel.com** (未加http://) or **www.deitelcom** (未加.com等組織單位))
 - 註: HTML5 **不會檢查**此網站是否可連通，只驗證網址格式

(少了1個點.)

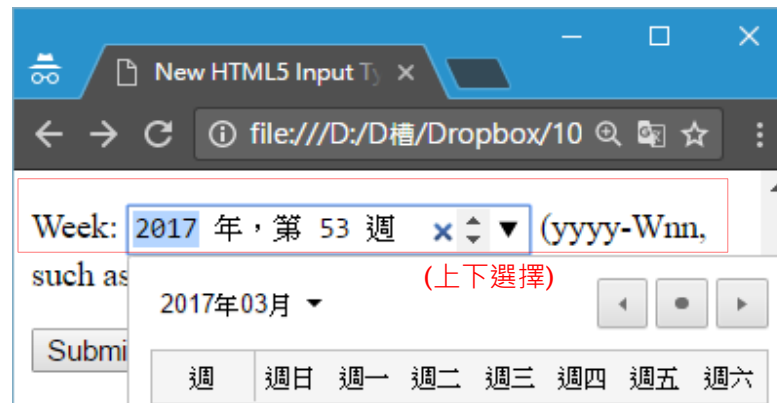


3.1.13 input Type week

(比如：孕婦懷孕週次)

`<input type = "week" />` 輸入週次 (ex: 2025-W01)

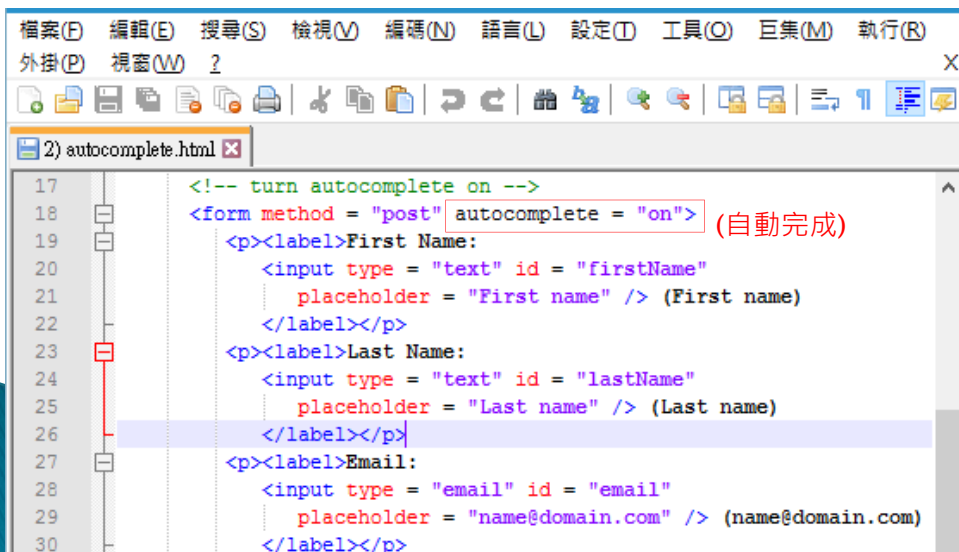
- ▶ To select a year and week number in the format `yyyy-Wnn` (`nn`是01-53週)
 - Example: 2025-w01代表2025第一週.
 - 註：除了Chrome、Opera，大多數瀏覽器顯示文字框



3.2 input and dataList Elements and autocomplete Attribute

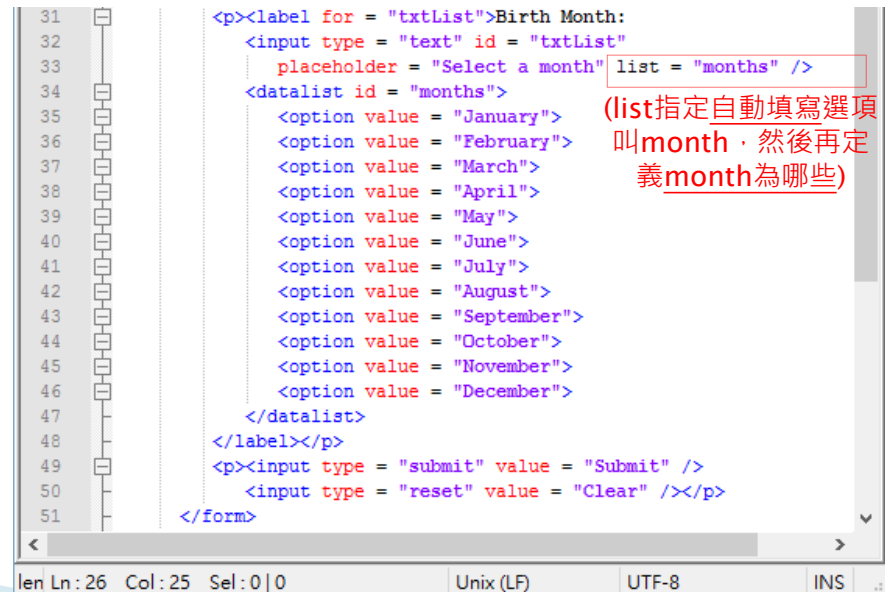
autocomplete (屬性：自動填寫)

- ▶ It shows **autocomplete** attribute and **dataList** element.
 - To automatically **fill** in the information based on **previous input**
 - 註：自動填寫過去曾輸入之資訊，如：name, address or e-mail.
- ▶ You can enable autocomplete for an entire form or just for **specific input elements**.
- ▶ **autocomplete = "on"**、**autocomplete = "off"**
 - 註：可自定哪些**input**要啟動“自動填寫”
 - 註：**name**、**email**適合自動填寫，但密碼、信用卡號則不適合



The screenshot shows a web browser window with a form titled "2) autocomplete.html". The form contains three input fields: "First Name", "Last Name", and "Email". The "First Name" field has a placeholder "First name" and the text "(First name)". The "Last Name" field has a placeholder "Last name" and the text "(Last name)". The "Email" field has a placeholder "name@domain.com" and the text "(name@domain.com)". The form is submitted using a "Submit" button. The "autocomplete" attribute is set to "on" for the entire form.

```
17 <!-- turn autocomplete on -->
18 <form method = "post" autocomplete = "on"> (自動完成)
19   <p><label>First Name:
20     <input type = "text" id = "firstName"
21       placeholder = "First name" /> (First name)
22   </label></p>
23   <p><label>Last Name:
24     <input type = "text" id = "lastName"
25       placeholder = "Last name" /> (Last name)
26   </label></p>
27   <p><label>Email:
28     <input type = "email" id = "email"
29       placeholder = "name@domain.com" /> (name@domain.com)
30   </label></p>
```

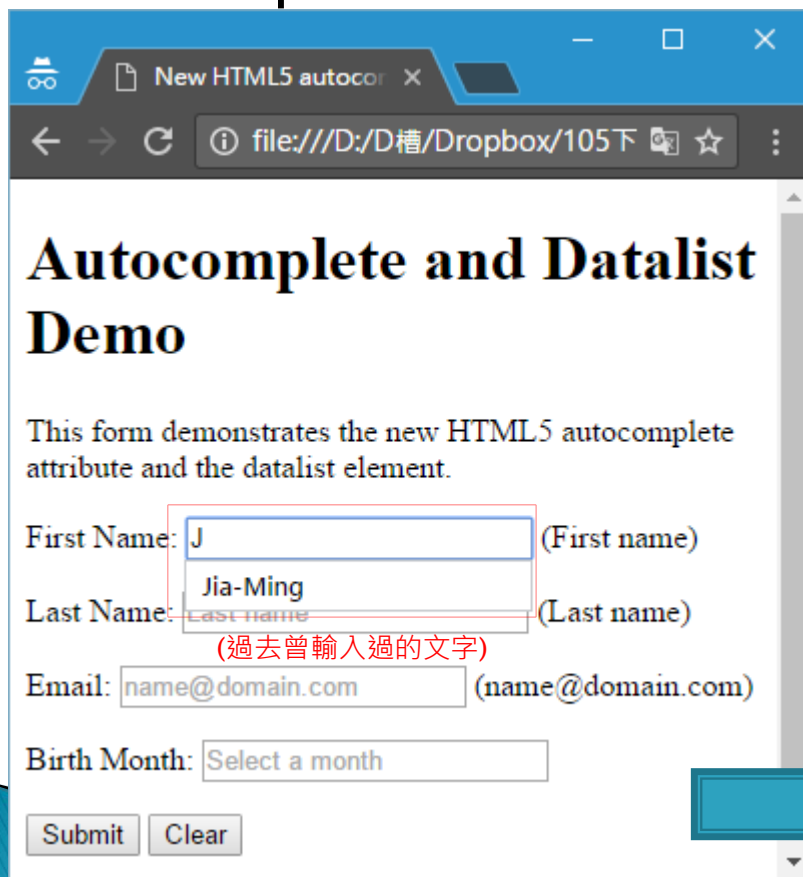


The screenshot shows a web browser window with a form titled "Birth Month". The form contains a dropdown menu with the placeholder "Select a month" and the text "(list指定自動填寫選項叫month，然後再定義month為哪些)". The dropdown menu is populated with the months of the year: January, February, March, April, May, June, July, August, September, October, November, and December. The form is submitted using a "Submit" button and a "Clear" button.

```
31 <p><label for = "txtList">Birth Month:
32   <input type = "text" id = "txtList"
33     placeholder = "Select a month" list = "months" />
34   <datalist id = "months">
35     <option value = "January">
36     <option value = "February">
37     <option value = "March">
38     <option value = "April">
39     <option value = "May">
40     <option value = "June">
41     <option value = "July">
42     <option value = "August">
43     <option value = "September">
44     <option value = "October">
45     <option value = "November">
46     <option value = "December">
47   </datalist>
48 </label></p>
49 <p><input type = "submit" value = "Submit" />
50   <input type = "reset" value = "Clear" /></p>
51 </form>
```

3.2 input and dataList Elements and autocomplete Attribute

- ▶ **autocomplete** (屬性：自動填完)
 - Example:



Autocomplete and Datalist Demo

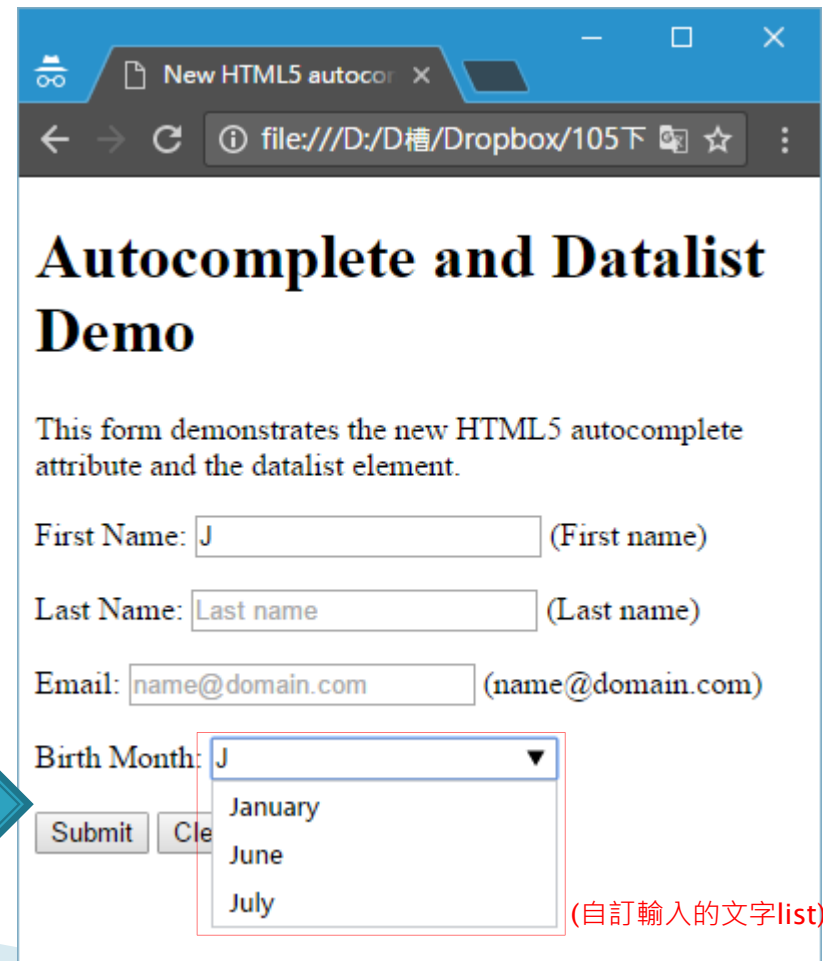
This form demonstrates the new HTML5 autocomplete attribute and the datalist element.

First Name: (First name)

Last Name: (Last name)

Email: (name@domain.com)

Birth Month:



Autocomplete and Datalist Demo

This form demonstrates the new HTML5 autocomplete attribute and the datalist element.

First Name: (First name)

Last Name: (Last name)

Email: (name@domain.com)

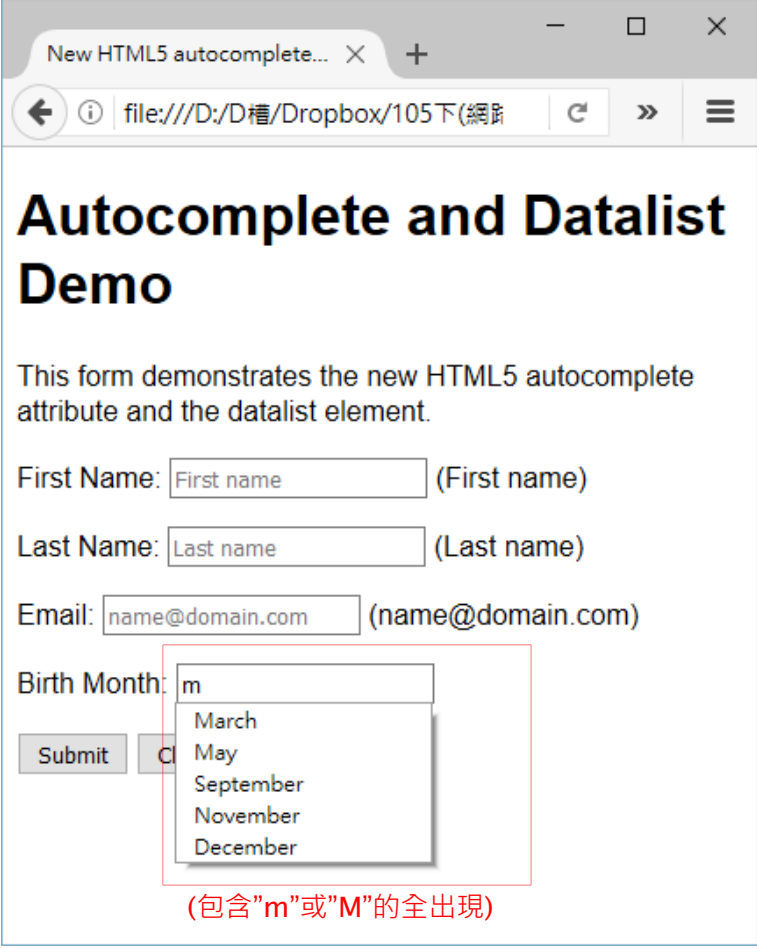
Birth Month: (Last name)

3.2.2 datalist Element

datalist 自訂填完選項

- ▶ To provide **input options** (輸入選項) for a **text input**.
- ▶ Ex: If the user types "J" in the text field, the list on months is narrowed to **January, June, and July**.
 - 註：輸入一個字母"J"，自動跳出"January, June, July"
- ▶ 註：Firefox will show the **drop-down list** (下拉選單) of months appears.
 - 註：Firefox若輸入"M"，則包含m或M的英文字都會出現，如：March, May, September, November及December.

Firefox 瀏覽器 (有些差異):



New HTML5 autocomplete... X +

file:///D:/D槽/Dropbox/105下(網路) < > ≡

Autocomplete and Datalist Demo

This form demonstrates the new HTML5 autocomplete attribute and the datalist element.

First Name: (First name)

Last Name: (Last name)

Email: (name@domain.com)

Birth Month:

Submit Clear

March
May
September
November
December

(包含"m"或"M"的全出現)



In-Class Exercise #1

課中練習：

- 表單中，請指定server接收網址: `action="https://120.114.190.22/web_course/get_post.php"`
- 用以檢視輸入結果

The screenshot shows a web browser window with a green title bar. The address bar displays a local file path: `file:///D:/D槽/Dropbox/105下(網路程式設計)/Slid`. The page content is a form titled "個人資訊調查".

Form fields and labels:

- 統計截止日期: 3/15
- 姓名: (必填)
- 學號:
- 出生日期: (yyyy-mm-dd)
- 家庭成員(數): (Enter a number between 0 and 5)
- 平均成績: 0 100
- 聯絡方式: (##) ###-####
- 聯絡信箱: (name@domain.com)
(灰字提示: placeholder)
- 個人網頁: (http://www.domainname.com)
- 幸運顏色: (Hexadecimal code such as #ADD8E6)

Buttons: Submit, Clear

(可參考ftp範例)

3.3 Page-Structure Elements

- HTML5針對文章(ex:新聞)有定義數種區域(area)：headers (標題欄), footers (頁尾), sections (章節), articles (內文), navigation areas (導覽區), asides (側邊), figures (圖片區), etc.



```
3) SectionElements.html
1  <!DOCTYPE html>
2
3  <!-- Fig. 3.18: sectionelements.html -->
4  <!-- New HTML5 section elements. -->
5
6  <html>
7    <head>
8      <meta charset="utf-8">
9      <title>New HTML5 Section Elements</title>
10    </head>
11
12    <body>
13      <header> <!-- header element creates a header for the page -->
14        <img src = "deitellogo.png" alt = "Deitel logo" />
15        <h1>Welcome to the Deitel Buzz Online</h1>
16
17        <!-- time element inserts a date and/or time -->
18        <time>2012-01-17</time>
19      </header>
20
21      <section id = "1"> <!-- Begin section 1 -->
22        <nav> <!-- nav element groups navigation links -->
23          <h2>Recent Publications</h2>
24          <ul>
25            <li><a href = "http://www.deitel.com/books/iw3http5">
26              Internet & World Wide Web How to Program, 5/e</a></li>
27            <li><a href = "http://www.deitel.com/books/androidfp">
28              Android for Programmers: An App-Driven Approach</a>
29            </li>
30            <li><a href = "http://www.deitel.com/books/iphonefp">
31              iPhone for Programmers: An App-Driven Approach</a></li>
32            <li><a href = "http://www.deitel.com/books/jhttp9">
33              Java How to Program, 9/e</a></li>
34            <li><a href = "http://www.deitel.com/books/cpphttp8">
35              C++ How to Program, 8/e</a></li>
36            <li>
37              <a href = "http://www.deitel.com/books/vcsharp2010http">
38                Visual C# 2010 How to Program, 4/e</a></li>
39            <li><a href = "http://www.deitel.com/books/vb2010http">
40              Visual Basic 2010 How to Program</a></li>
41          </ul>
42        </nav>
43      </section>
```

```
45
46 <section id = "2"> <!-- Begin section 2 -->
47   <h2>How to Program Series Books</h2>
48   <h3><em>Java How to Program, 9/e</em></h3>
49
50   <figure> <!-- figure element describes the image -->
51     <img src = "jhttp.jpg" alt = "Java How to Program, 9/e" />
52
53     <!-- figurecaption element inserts a figure caption -->
54     <figcaption><em>Java How to Program, 9/e</em>
55       cover.</figcaption>
56   </figure>
57
58   <!-- article element represents content from another source -->
59   <article>
60     <header>
61       <h5>From
62       <em>
63         <a href = "http://www.deitel.com/books/jhttp9/">
64           Java How to program, 9/e: </a>
65       </em>
66     </h5>
67   </header>
68
69   <p>Features include:
70   <ul>
71     <li>Rich coverage of fundamentals, including
72       <!-- mark element highlights text -->
73       <mark>two chapters on control statements.</mark></li>
74     <li>Focus on <mark>real-world examples.</mark></li>
75     <li><mark>Making a Difference exercises set.</mark></li>
76     <li>Early introduction to classes, objects,
77     <li>Integrated exception handling.</li>
78     <li>Files, streams and object serialization.</li>
79     <li>Optional modular sections on language and
80     <li>Other topics include: Recursion, searching,
81   </ul>
```

Ps. 使用標籤，方便套用CSS樣式 (下一章會教)

3.3 Page-Structure Elements

- ▶ (接續) HTML5針對文章(ex:新聞) 有定義數種區域(area)：headers, footers, sections, articles, navigation areas, asides, figures, etc.



```
91 <details>
92 <summary>Recent Edition Testimonials</summary>
93 <ul>
94 <li>"Updated to reflect the state of the
95 art in Java technologies; its deep and
96 crystal clear explanations make it
97 indispensable. The social-consciousness
98 [Making a Difference] exercises are
99 something really new and refreshing."
100 <strong>&mdash;Joseacute; Antonio
101 Gonzacute;lez Seco, Parliament of
102 Andalusia</strong></li>
103 <li>"Gives new programmers the benefit of the
104 wisdom derived from many years of software
105 development experience."<strong>
106 &mdash;Edward F. Gehringer, North Carolina
107 State University</strong></li>
108 <li>"Introduces good design practices and
109 methodologies right from the beginning.
110 An excellent starting point for developing
111 high-quality robust Java applications."
112 <strong>&mdash;Simon Ritter,
113 Oracle Corporation</strong></li>
114 <li>"An easy-to-read conversational style.
115 Clear code examples propel readers to
116 become proficient in Java."
117 <strong>&mdash;Patty Kraft, San Diego State
118 University</strong></li>
119 <li>"A great textbook with a myriad of examples
120 from various application domains&mdash;
121 excellent for a typical CS1 or CS2 course."
122 <strong>&mdash;William E. Duncan, Louisiana
123 State University</strong></li>
124 </ul>
125 </details>
126 </p>
127 </article>
128
129 <!-- aside element represents content in a sidebar that's -->
130 <!-- related to the content around the element -->
131 <aside>
132 The aside element is not formatted by the browsers.
133 </aside>
```

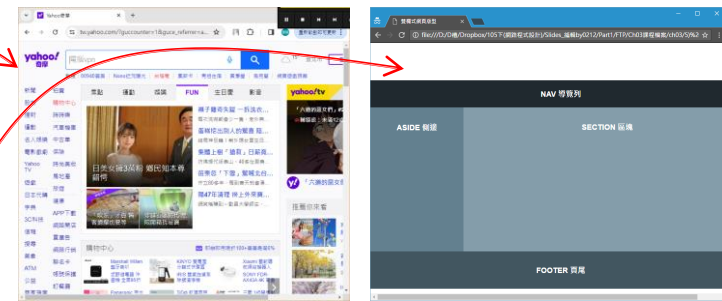
(article)
內文

```
130 <!-- related to the content around the element -->
131 <aside>
132 The aside element is not formatted by the browsers.
133 </aside>
134
135 <h2>Deitel Developer Series Books</h2>
136 <h3><em>Android for Programmers: An App-Driven Approach
137 </em></h3>
138 Click <a href = "http://www.deitel.com/books/androidfp/">
139 here</a> for more information or to order this book.
140
141 <h2>LiveLessons Videos</h2>
142 <h3><em>C# 2010 Fundamentals LiveLessons</em></h3>
143 Click <a href = "http://www.deitel.com/Books/LiveLessons/">
144 here</a> for more information about our LiveLessons videos.
145 </section>
146
147 <section id = "3"> <!-- Begin section 3 -->
148 <h2>Results from our Facebook Survey</h2>
149 <p>If you were a nonprogrammer about to learn Java for the first
150 time, would you prefer a course that taught Java in the
151 context of Android app development? Here are the results from
152 our survey:</p>
153
154 <!-- meter element represents a scale within a range -->
155 0 <meter min = "0"
156 max = "54"
157 value = "14"></meter> 54
158 <p>Of the 54 responders, 14 (green) would prefer to
159 learn Java in the context of Android app development.</p>
160 </section>
161
162 <!-- footer element represents a footer to a section or page, -->
163 <!-- usually containing information such as author name, -->
164 <!-- copyright, etc. -->
165 <footer>
166 <!-- wbr element indicates the appropriate place to break a -->
167 <!-- word when the text wraps -->
168 <h6>&copy; 1992-2012 by Deitel &amp; Associ&wbr;ates, Inc.
169 All Rights Reserved.</h6>
170 <!-- address element represents contact information for a -->
171 <!-- document or the nearest body element or article -->
172 <address>
173 Contact us at <a href = "mailto:deitel@deitel.com">
174 deitel@deitel.com</a>
175 </address>
176 </footer>
177 </body>
```

(aside)
側邊

(footer註腳)

3.3.1 header Element



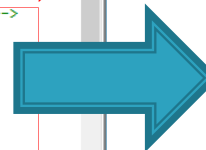
<header> ... </header> 網頁文章標題 (可包含：圖/文)

- ▶ **標題**：Creates a **header** for this page that contains both **text** and **graphics**.
 - Can be used **multiple times** (可多次使用) on a page
 - Can include HTML headings (**<h1>** through **<h6>**), navigation (導覽連結), **images** (圖) and **logos** (商標).

<time> ... </time> 時間描述

- 註：不限於header，用於時間描述

```
3) SectionElements.html
1 <!DOCTYPE html>
2
3 <!-- Fig. 3.18: sectionelements.html -->
4 <!-- New HTML5 section elements. -->
5 <html>
6 <head>
7 <meta charset="utf-8">
8 <title>New HTML5 Section Elements</title>
9 </head>
10
11 <body>
12 <header> <!-- header element creates a header for the page -->
13 <img src = "deitellogo.png" alt = "Deitel logo" />
14 <h1>Welcome to the Deitel Buzz Online</h1>
15
16 <!-- time element inserts a date and/or time -->
17 <time>2012-01-17</time>
18 </header>
19
20
21 <section id = "1"> <!-- Begin section 1 -->
22 <nav> <!-- nav element groups navigation links -->
23 <h2>Recent Publications</h2>
24 <ul>
25 <li><a href = "http://www.deitel.com/books/iw3http5">
26 Internet & World Wide Web How to Program, 5/e</a></li>
27 <li><a href = "http://www.deitel.com/books/androidfp">
28 Android for Programmers: An App-Driven Approach</a>
29 </li>
30 <li><a href = "http://www.deitel.com/books/iphonefp">
31 iPhone for Programmers: An App-Driven Approach</a></li>
32 <li><a href = "http://www.deitel.com/books/jhtp9">
33 Java How to Program, 9/e</a></li>
34 <li><a href = "http://www.deitel.com/books/cnphttp8">
```



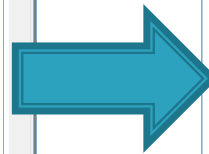
3.3.2 nav Element

`<nav> ... </nav>` 網頁導覽連結

- ▶ The **nav element** groups navigation links.
 - Example: 下例用Recent Publications及 7個 **ul** 無序條列連結到其他導覽分頁。

(nav導覽連結)

```
<section id = "1"> <!-- Begin section 1 -->
<nav> <!-- nav element groups navigation links -->
<h2> Recent Publications</h2>
<ul>
<li><a href = "http://www.deitel.com/books/iw3htp5">
    Internet & World Wide Web How to Program, 5/e</a></li>
<li><a href = "http://www.deitel.com/books/androidfp/">
    Android for Programmers: An App-Driven Approach</a>
</li>
<li><a href = "http://www.deitel.com/books/iphonefp">
    iPhone for Programmers: An App-Driven Approach</a></li>
<li><a href = "http://www.deitel.com/books/jhtp9/">
    Java How to Program, 9/e</a></li>
<li><a href = "http://www.deitel.com/books/cpphtp8/">
    C++ How to Program, 8/e</a></li>
<li>
    <a href = "http://www.deitel.com/books/vcsharp2010htp">
        Visual C# 2010 How to Program, 4/e</a></li>
<li><a href = "http://www.deitel.com/books/vb2010htp">
        Visual Basic 2010 How to Program</a></li>
</ul>
</nav>
</section>
```

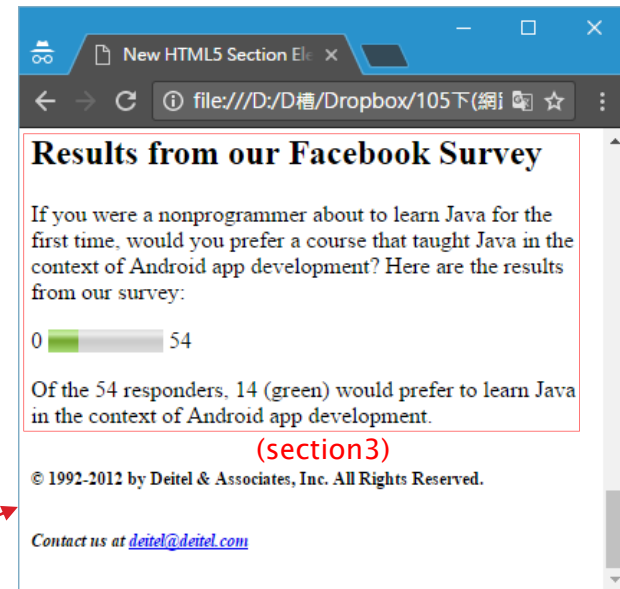
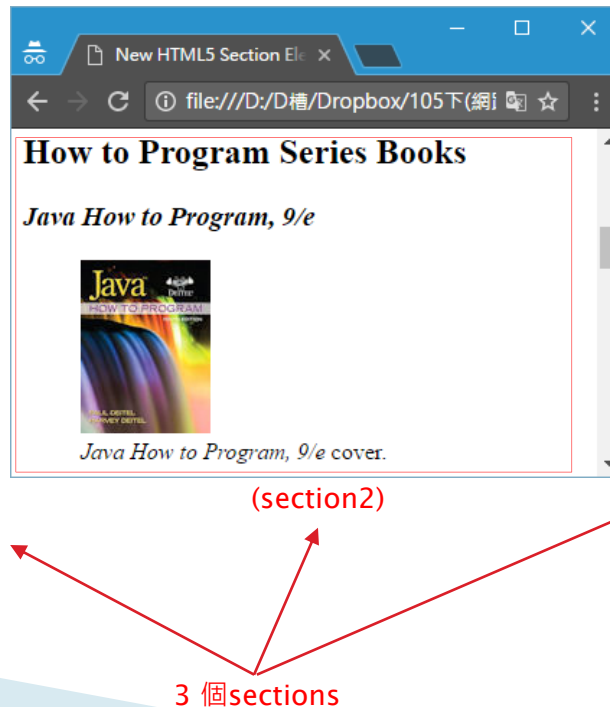


(導覽到其他分頁的連結)

3.3 Page-Structure Elements

`<section id=...>...</section>` 文章章節

- ▶ Describes a section of a document (文章中的一章節)
 - 註：通常每個章會有一個標題，階層性呈現
 - Example: This document has three sections—
 - The first is Recent Publications.
 - The section element may also be nested (巢狀/階層狀) in an article.



3 個sections

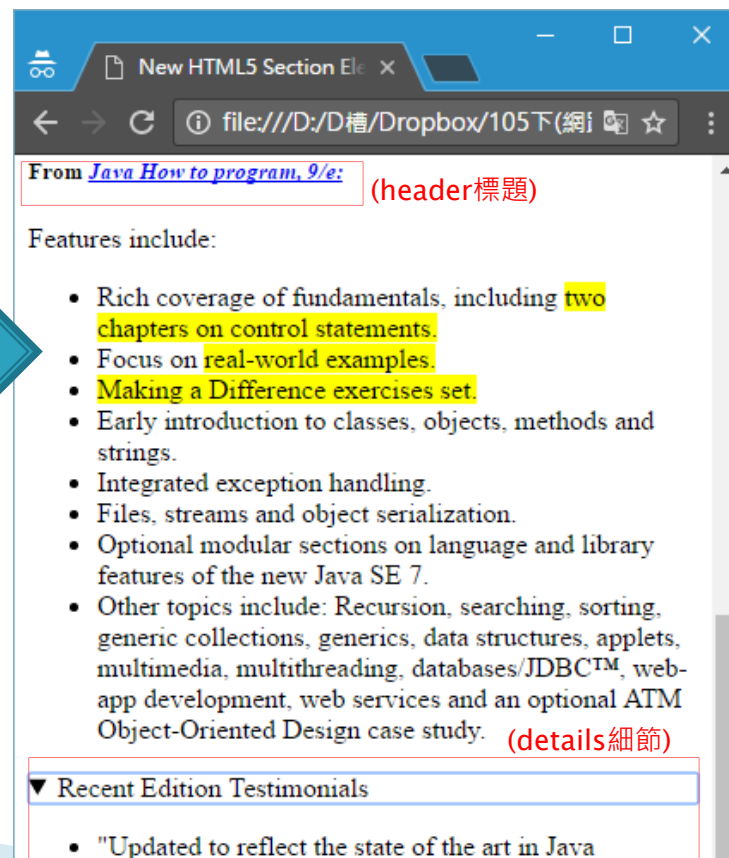
3.3.4 article Element

(例如：網誌)

`<article> ... </article>` 內文 (可含多個章節)

- ▶ Describes standalone content (獨立內容) that could potentially be used or distributed (發佈) elsewhere, such as a news article (新聞訊息), forum post (論壇文章) or blog entry (網誌).
 - 包含結構：`<header> ... </header>` 文章標題、`<details> ... </details>` 文章細節。

```
58 <article>
59 <header>
60 <h5>From
61 <em>
62 <a href = "http://www.deitel.com/books/jhtp9/">
63 Java How to program, 9/e: </a>
64 </em>
65 </h5>
66 </header>
67
68 <p>Features include:
69 <ul>
70 <li>Rich coverage of fundamentals, including
71 <!-- mark element highlights text -->
72 <mark>two chapters on control statements.</mark></li>
73 <li>Focus on <mark>real-world examples.</mark></li>
74 <li><mark>Making a Difference exercises set.</mark></li>
75 <li>Early introduction to classes, objects,
76 <li>Integrated exception handling.</li>
77 <li>Files, streams and object serialization.</li>
78 <li>Optional modular sections on language and
79 <li>Other topics include: Recursion, searching,
80 </ul>
81
82 (header文章標題)
83
84 (detail細節)
```



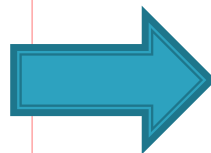
3.3.5 summary Element and details Element

`<summary>...</summary>` 文章摘要

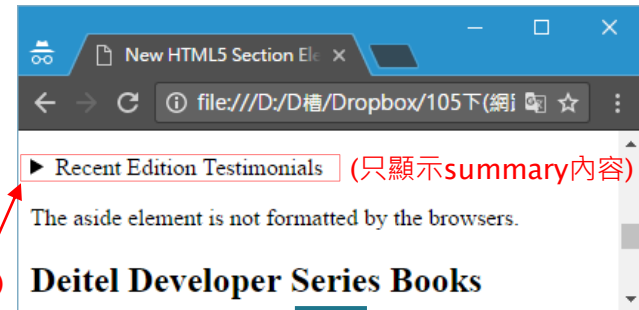
- ▶ 註：呈現 **right-pointing arrow** (向右箭頭) next to a summary or caption when the document is rendered in a browser (Fig. 3.19).

- 註：用滑鼠點後，箭頭變向下，並展示 **details element**
`<details>...</details>`.

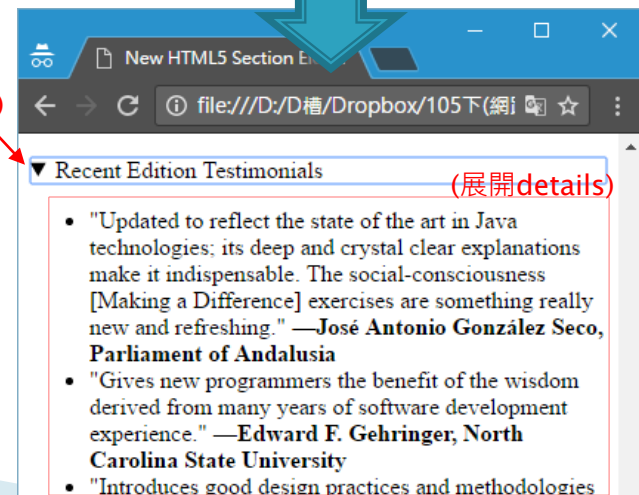
```
91  (details) <details>
92  <summary>Recent Edition Testimonials</summary> (summary)
93  <ul>
94  <li>"Updated to reflect the state of the
95  art in Java technologies; its deep and
96  crystal clear explanations make it
97  indispensable. The social-consciousness
98  [Making a Difference] exercises are
99  something really new and refreshing."
100 <strong>&mdash;Jos&eacute; Antonio
101 Gonz&aacute;lez Seco, Parliament of
102 Andalusia</strong></li>
103 <li>"Gives new programmers the benefit of the
104 wisdom derived from many years of software
105 development experience."<strong>
106 &mdash;Edward F. Gehringer, North Carolina
107 State University</strong></li>
108 <li>"Introduces good design practices and
109 methodologies right from the beginning.
110 An excellent starting point for developing
111 high-quality robust Java applications."
112 <strong>&mdash;Simon Ritter,
113 Oracle Corporation</strong></li>
114 <li>"An easy-to-read conversational style.
115 Clear code examples propel readers to
116 become proficient in Java."
117 <strong>&mdash;Patty Kraft, San Diego State
118 University</strong></li>
119 <li>"A great textbook with a myriad of examples
120 from various application domains&mdash;
121 excellent for a typical CS1 or CS2 course."
122 <strong>&mdash;William E. Duncan, Louisiana
123 State University</strong></li>
124 </ul>
125 </details>
```



(三角形向右)



(三角形向下)



3.3.3 figure Element and figcaption Element

(建立標籤，後面較方便套用CSS樣式)

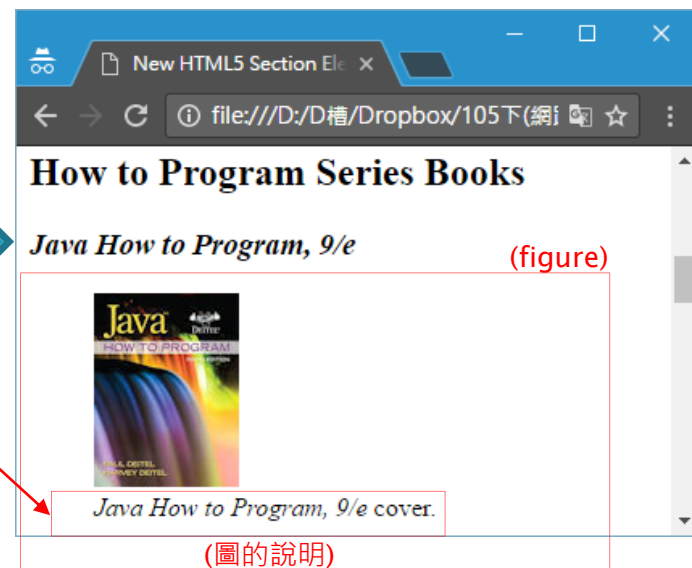
`<figure> ... </figure>` 圖形區塊

- ▶ To describe a figure (e.g., an image/圖像, chart or table/圖表).

`<figcaption> ... </figcaption>` 圖的說明

- ▶ Provides a caption (說明) for the image in the figure element.

```
45 <section id = "2"> <!-- Begin section 2 -->
46 <h2>How to Program Series Books</h2>
47 <h3><em>Java How to Program, 9/e</em></h3>
48
49 <figure> <!-- figure element describes the image -->
50 <img src = "jhttp.jpg" alt = "Java How to Program, 9/e" />
51
52 <!-- figurecaption element inserts a figure caption -->
53 <figcaption><em>Java How to Program, 9/e</em>
54 cover.</figcaption>
55 </figure>
56
57 <!--article element represents content from another source -->
58 <article>
59 <header>
60 <h5>From
61 <em>
62 <a href = "http://www.deitel.com/books/jhttp9/">
63 Java How to program, 9/e: </a>
64 </em>
65 </h5>
66 </header>
```

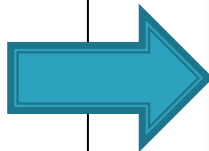


3.3.7 aside Element

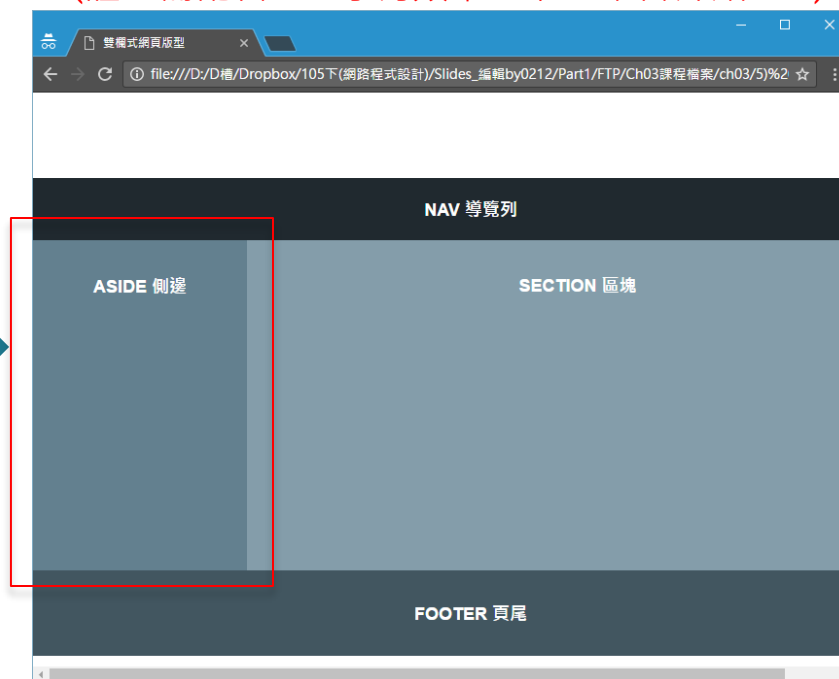
`<aside>...</aside>` 側邊區塊

- ▶ Describes **content** that's related to the **surrounding content** but separate from the flow of the text.
 - 如：放新聞文章的背景歷史。

```
5) aside.html x
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <meta http-equiv="Content-Type" content="text/html">
5   <title>雙欄式網頁版型</title>
6
7   <style type="text/css">
58 </style>
59 </head>
60 <body>
61 <header id="header">
62   <h1>HEADER 標題</h1>
63 </header>
64
65 <nav id="nav">
66   <h1>NAV 導覽列</h1>
67 </nav>
68
69 <aside id="aside">
70   <h1>ASIDE 側邊</h1>
71 </aside>
72
73 <section id="section">
74   <h1>SECTION 區塊</h1>
75 </section>
76
77 <footer id="footer">
78   <h1>FOOTER 頁尾</h1>
79 </footer>
80
81 </body>
82 </html>
```



(註：需配合CSS才有效果，下一章會介紹CSS)



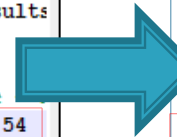
3.3.8 meter Element

- `<meter ...> ... </meter>` 計量器 (視覺化呈現數值)
- ▶ a visual representation (視覺化呈現) of a measure within a range (一段範圍).
 - 範例: 統計值應用 (ex: 54個人回應，其中14有興趣學習android app)
 - `min="..."`: 範圍最小值
 - `max="..."`: 範圍最大值.
 - `value="..."`: 呈現的數值.

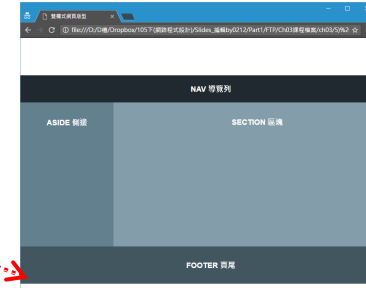
```

147 <section id = "3"> <!-- Begin section 3 -->
148 <h2>Results from our Facebook Survey</h2>
149 <p>If you were a nonprogrammer about to learn Java for the
150 time, would you prefer a course that taught Java in the
151 context of Android app development? Here are the results
152 our survey:</p>
153
154 <!-- meter element represents a scale within a range (meter)
155 0 <meter min = "0" max = "54" value = "14"></meter> 54
156
157 <p>Of the 54 responders, 14 (green) would prefer to
158 learn Java in the context of Android app development.</p>
159 </section>
160

```



3.3.9 footer Element



<footer>...</footer> 頁尾 (通常放置：聯繫方式、地址)

► Describes a *footer* (頁尾)

- usually appears at the bottom (最底部) of the page or section element.
 - 註：網頁最後或section下方
- Example: the footer describes the copyright notice (著作權) and contact information (聯繫資訊).

```
157 <p>Of the 54 responders, 14 (green) would prefer to
158 learn Java in the context of Android app development.</p>
159 </section>
160
161 <!-- footer element represents a footer to a section or page,
162 <!-- usually containing information such as author name, -->
163 <!-- copyright, etc. --> (footer頁尾)
164 <footer>
165 <!-- wbr element indicates the appropriate place to break
166 <!-- word when the text wraps -->
167 <h6>&copy; 1992-2012 by Deitel & Associates, Inc.
168 All Rights Reserved.</h6>
169 <!-- address element represents contact information for a
170 <!-- document or the nearest body element or article -->
171 <address>
172 Contact us at <a href = "mailto:deitel@deitel.com">
173 deitel@deitel.com</a>
174 </address>
175 </footer>
176 </body>
```



3.3.10 (補充)Text-Level Semantics: mark Element and wbr Element

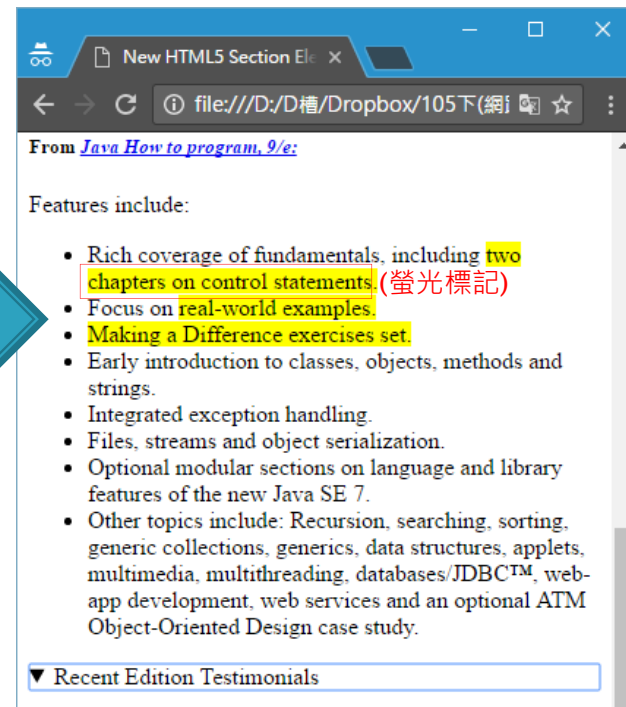
`<mark>...</mark>` 螢光標示(重點)

- ▶ To highlight (強調) the text.

`<wbr>` word breaker

- ▶ 註：當文字較長時，文字可斷行之處（避免排版不整齊）。

```
58 <article>
59 <header>
60 <h5>From
61 <em>
62 <a href = "http://www.deitel.com/books/jhttp9/">
63   Java How to program, 9/e: </a>
64 </em>
65 </h5>
66 </header>
67
68 <p>Features include:
69 <ul>
70 <li>Rich coverage of fundamentals, including
71 <!-- mark element highlights text --> (螢光標記)
72 <mark>two chapters on control statements.</mark></li>
73 <li>Focus on <mark>real-world examples.</mark></li>
74 <li><mark>Making a Difference exercises set.</mark></li>
75 <li>Early introduction to classes, objects,
76 <li>Integrated exception handling.</li>
77 <li>Files, streams and object serialization.</li>
78 <li>Optional modular sections on language and
79 <li>Other topics include: Recursion, searching,
80 </ul>
81
82
```



In-Class Exercise #2

- ▶ Create an **HTML5 document** that contains **nav, header, article, section, footer, mark, summary, detail**.
- ▶ 課中練習：請同學依下列建立一個html5文件，包含有**nav, header, article, section, footer, mark, summary, detail**.

The screenshot shows a web browser window with the address bar displaying `file:///D:/D槽/Dropbox/105下(網路)`. The page content is annotated with red boxes and labels:

- Navigation Bar:** `<nav>` 導覽區塊 (Annotations: 首頁, Yahoo新聞, Facebook)
- Header:** `<header>` 標題欄 (Content: **** MIS2000 Lab. ****)
- Article:** `<article>` 內文 (Content: 自我介紹與出版書籍, 關於MIS2000 Lab.)
- Section:** `<section>` 章節 (Content: 自我介紹, marked with a yellow highlight and "螢光標示")
- Summary:** `<summary>` 摘要 (Content: 簡單的摘要說明。關於MIS2000 Lab.)
- Details:** `<details>` 詳細內容 (Content: 出版書籍, followed by a list of 4 items)
- Footer:** `<footer>` 頁尾 (Content: Last updated on August 01,)



Lab

(上機練習)



In-Class Exercise #1

課中練習：

- 請指定server接收網址，以驗證輸入內容：

`action=" https://120.114.190.22/web_course/get_post.php"`

The screenshot shows a web browser window with a green title bar and a dark address bar. The address bar contains the file path: `file:///D:/D槽/Dropbox/105下(網路程式設計)/Slid`. The main content area displays a form titled "個人資訊調查". The form includes the following fields and labels:

- 統計截止日期：3/15
- 姓名: (必填)
- 學號:
- 出生日期: (yyyy-mm-dd)
- 家庭成員(數): (Enter a number between 0 and 5)
- 平均成績: 0 100
- 聯絡方式: (##) ###-####
- 聯絡信箱: (name@domain.com)
- 個人網頁: (http://www.domainname.com)
- 幸運顏色: (Hexadecimal code such as #ADD8E6)

At the bottom of the form are two buttons: "Submit" and "Clear". A red text annotation "(灰字提示：placeholder)" is placed over the email field.

In-Class Exercise #2

- ▶ Create an **HTML5 document** that contains **nav, header, article, section, footer, mark, summary, detail**.
- ▶ 課中練習：請同學依下列建立一個html5文件，包含有**nav, header, article, section, footer, mark, summary, detail**.

The screenshot shows a web browser window with the following content and annotations:

- Navigation Bar:** 首頁, Yahoo新聞, Facebook. Labeled with **<nav> 導覽**.
- Header:** ** MIS2000 Lab. **. Labeled with **<header> 標頭**.
- Section:** 自我介紹. Labeled with **<section>**.
- Article Content:** 關於MIS2000 Lab. Labeled with **<article> 內文**.
- Summary:** 出版書籍. Labeled with **<summary>**.
- Detail:** 自我介紹. Labeled with **<detail>**.
- Footer:** Last updated on August 01, 2014. Labeled with **<footer>**.

- 注意：

今日 23:59 前上傳到：

<ftp://111webdesign:webdesign@120.114.190.22:4000>

- Lab03
- 請上傳至“學號”資料匣 (否則不計分)



【好文章分享】



找到適合自己的mentor (益友)

► [摘錄自：商業週刊]

https://www.businessweekly.com.tw/careers/blog/3009281?fbclid=IwAR2UOue-MER6OM3hTRX0ZTpdlgrAf5_s5Y9avli0sOhiwMMM4DtVa5cAXcw

以前在美國工作的時候，剛進入公司，主管就找了一個資深同事當我的mentor，他不只給了我專業的指導，也給我職場上處事的建議。

後來，我也找了幾位前輩，與他們建立mentor關係，每次工作遇到瓶頸，他們都是那個指引我的角色。創業以後，更是需要mentor，在每一次需要找人討論的時候，他們就是那個最可靠的諮詢對象。

職場導師（mentor）在美國是個蠻流行的概念，在台灣似乎比較不流行，有些人覺得主管就是這個角色，但其實並不完全相同。主管和下屬有著特殊關係，你的個人職涯未必是主管最在意的事。但mentor是那個和你沒有利害關係，願意跟你說真話的人，因此會很建議每個人在職場都有幾個mentor，讓你變成更好的自己。